2025 CpSc H Q2

Section: Computer Systems

Topic: Security Risks, Precautions and Legislation — Digital Signatures

Question Summary

State two benefits of using a digital signature included with a document e-mailed by a bank.

Worked Solution

Authenticity (Identity Assurance): A valid signature proves the message was signed by the claimed sender because only the sender has the private key to create that signature.

Integrity (Tamper Detection): If the signed document is altered in any way after signing, signature verification fails, revealing that it has been changed.

Optional additional benefit (useful to remember): Non-repudiation — the sender cannot credibly deny having sent the signed document later.

Why these work (brief cryptography intuition)

Digital signatures use public-key cryptography. The sender signs a hash of the document with their private key. Anyone with the sender's public key can verify the signature matches the document's hash; any change breaks the match.

Final Answer

Two correct benefits are: Authenticity and Integrity. (Non-repudiation is also acceptable.)

Revision Tips

• For two's complement and digital-security questions on the same page, keep answers separate and concise.

• Remember: digital signature ≠ encryption. Encryption hides content; signatures prove identity and detect changes.

- Use the exact terms the SQA expects: Authenticity; Integrity; (optionally) Non-repudiation.